DHRUBJYOTI SHARMA

Ph.D. Scholar, Biological Sciences and Engineering - Indian Institute of Technology Gandhinagar

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RESEARCH SUMMARY

Ph.D. candidate in Structural Biology with expertise spanning experimental and computational methods for characterizing disease-associated proteins. Current research focuses on the structure and function of a therapeutic target protein implicated in Alzheimer's disease and its interaction with amyloid precursor protein (APP), combining cryo-electron microscopy, X-ray crystallography, biophysical characterization (ITC, FPLC/SEC), and computational drug discovery (molecular docking, molecular dynamics, MM-GBSA, ADME profiling). Experienced in solving cryo-EM structures end-to-end, from data collection troubleshooting through refinement and in running iterative, hypothesis-driven computational-to-experimental research loops.

EDUCATION

Ph.D., Structural Biology / Molecular Biology / Computational Biology *Indian Institute of Technology Gandhinagar •*

2022 – Present

Investigating the structure and function of a disease-associated protein target and its interaction with APP in Alzheimer's disease, using integrated computational and wet-lab approaches.

M.Sc., Biotechnology *SRM Institute of Science and Technology • 2019*

B.Sc., Biotechnology *Assam University • 2017*

SKILLS

Wet Lab:

Protein purification (His-tagged), protein crystallization, SDS-PAGE, Western blotting, molecular cloning, mammalian cell culture, FPLC/SEC, Circular Dichroism, Isothermal Titration Calorimetry (ITC / PEAQ-ITC), CellTiter-Glo viability assays, ELISA-based functional assays.

Computational - Structural & Drug Discovery:

Schrödinger Suite (Glide XP/SP docking, Desmond MD simulations up to 1000 ns, Prime MM-GBSA, Prime Induced Fit Docking, QikProp ADME prediction, Jaguar QM/MM and MEP/HOMO-LUMO analysis, Maestro), AutoDock, GROMACS, PyMOL, ChimeraX, integrative structural modeling (IMP), trajectory analysis (MDAnalysis).

Structural & Imaging Techniques:

Cryo-EM data processing (cryoSPARC — particle picking, 2D/3D classification, 3D variability analysis, preferred-orientation mitigation), map post-processing (DeepEMhancer), model refinement (Phenix real-space refinement), X-ray diffraction data processing, negative-stain EM.

Programming & Data:

Python for structural bioinformatics and data processing/analysis scripting.

RESEARCH EXPERIENCE

Doctoral Research Scholar *• 2022 – Present*

- Lead structural and biophysical characterization of a therapeutic target protein implicated in Alzheimer's disease, integrating cryo-EM, X-ray crystallography, and biochemical assays across three thesis objectives.
- Solved a novel cryo-EM structure (homodimer, ~4.77 Å, C2 symmetry) using cryoSPARC, resolving preferred-orientation bias through iterative template picking, re-picking, and 3D variability analysis; performed downstream refinement with DeepEMhancer and Phenix real-space refinement.
- Designed and executed ITC binding studies characterizing target protein dimerization and small-molecule interactions, including thermogram troubleshooting and quantitative analysis in PEAQ-ITC.
- Ran end-to-end computational drug discovery workflows in the Schrödinger Suite - Glide XP docking, Desmond MD, Prime MM-GBSA, and QikProp ADME prediction - across two independent inhibitor discovery programs, with results incorporated into manuscripts in preparation.

- Built an integrative structural model of a multi-protein membrane complex using IMP, encoding experimentally derived interaction restraints into PMI topology scripts.
- Validated a disease-relevant protein's dimerization mechanism via SDS-PAGE/Western blotting, confirming a cysteine-dependent disulfide-linked dimer.
- Mentored and guided 20 interns/trainees across wet-lab and computational research projects during doctoral studies, including experimental design, technique training, and data interpretation

Graduate Teaching Assistant — Structural Biology Practicals *IIT Gandhinagar • 3 semesters*

- Designed and delivered laboratory practicals for structural biology coursework, including development of a model-protein-based (HEWL) 14-week practical curriculum.

Research Scholar *Bioinformatics Centre, Dept. of Biotechnology (Govt. of India), Gurucharan College, Silchar, Assam*

Jan–Jun 2020

- Developed and maintained a bioinformatics database of disease biomarkers.
- Supported infrastructural development, resource management, and guidance for traineeship/studentship programs.

Lecturer, Department of Biotechnology *Gurucharan College, Silchar, Assam • Oct 2020 - Apr 2021*

- Supervised undergraduate (TDC Science Stream) research projects.

PUBLICATIONS

Sharma, D. et al. Structural and biophysical characterization of a disease-relevant protein dimerization mechanism. Manuscript under review

Patel, M., Aghara, H., Chadha, P., Solanki, H., **Sharma, D**., Thiruvenkatam, V., & Mandal, P. (2026). Synergistic Effect of Sodium Acetate and Sodium Butyrate in Ameliorating Ethanol-Induced Hepatic Inflammation Through Modulation of the NF-κB Signaling Pathway. Mediators of Inflammation, 2026(1), 5835567.

Chadha, P., Aghara, H., Solanki, H., Patel, M., **Sharma, D**., Thiruvenkatam, V., & Mandal, P. (2025). Modulation of TNFα-driven neuroinflammation by Gardenin A: insights from in vitro, in vivo, and in silico studies. Frontiers in Pharmacology, 16, 1681403.

Chadha, P., Aghara, H., Johnson, D., **Sharma, D**., Odedara, M., Patel, M., Kumar, H., Thiruvenkatam, V. and Mandal, P. (2025). Gardenin A alleviates alcohol-induced oxidative stress and inflammation in HepG2 and Caco2 cells via AMPK/Nrf2 pathway. Bioorganic Chemistry, 161, 108543.

Naha, A., Debroy, R., **Sharma, D**., Shah, M.P. and Nath, S. (2023). Microbial fuel cell: A state-of-the-art and revolutionizing technology for efficient energy recovery. Cleaner and Circular Bioeconomy, 100050.

Sannasimuthu, A., **Sharma, D**., Paray, B.A., Al-Sadoon, M.K. and Arockiaraj, J. (2020). Intracellular oxidative damage due to antibiotics on gut bacteria reduced by glutathione oxidoreductase-derived antioxidant molecule GM15. Archives of Microbiology, 1–7.

CONFERENCES & PRESENTATIONS

- Oral Presentation (accepted), IUCr 2026 - International Union of Crystallography Congress, Calgary, Canada (August 2026).
- Presented Poster, “Horizons in Structural and Computational Biology, 2025”, February 28 & March 01, 2025
- Participant, “Understanding the Breathing of Biomolecules: Recent Advances in Cryo-EM and Chemical Biology,” IIT Bombay (March 2024).
- 6-day workshop, “Drug Design Against COVID-19 Targets,” Wisecorner Laboratories & CIIC, Chennai (2020).
- Workshop, “Python for Biologists” (2020); Workshop, “Targeted Gene Editing using CRISPR Technology” (2020).
- 15-day training, Guwahati Biotech Park, IIT Guwahati (2018).

ACHIEVEMENTS & CERTIFICATIONS

- Accepted Oral presentation, IUCr 2026, Calgary (2026).
- Scientific Writing Certification Course, IIT Gandhinagar (2024).
- Qualified GATE 2022, Biotechnology (AIR 1323).
- Qualified CSIR NET (Lectureship), Life Science (AIR 96).